Confusion Matrix

A confusion matrix summarises classification results as counts of true positives, false positives, true negatives, and false negatives. Every other classification metric is derived from these four numbers.

Accuracy is the proportion of correct predictions overall. It is misleading on imbalanced datasets: a classifier that always predicts the majority class can reach very high accuracy while being useless.

Precision, Recall and F1

Precision is the fraction of predicted positives that are actually positive, so it answers the question of how much we can trust a positive prediction. Recall is the fraction of actual positives that the model found, so it answers how much of the target class we are missing.

The F1 score is the harmonic mean of precision and recall. The harmonic mean is used rather than the arithmetic mean because it punishes a large imbalance between the two values.

ROC and AUC

The ROC curve plots the true positive rate against the false positive rate as the classification threshold is varied. A model that ranks all positives above all negatives traces a curve through the top-left corner.

The area under the ROC curve, or AUC, summarises the curve as a single number and can be interpreted as the probability that a randomly chosen positive is ranked above a randomly chosen negative.